Muhammad Muizz Zafar

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Security Clearance

Reliability Level Security Clearance May 2024

Skills

Programming Languages:	Java, Python, GoLang, JavaScript, Kotlin, C#, SQL, HTML/CSS, PHP, Flutter, C++
Tools/Frameworks:	AWS, Splunk, Flask, React, Vue, Angular, GraphQl, Docker, Git, Firebase, Linux

Education

University of Ottawa Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* Sept 2020 - Dec 2024

Work experience

Software & Database Analyst May 2024 - Present
Elections Canada Ottawa, ON

- Built and maintained scalable **ETL pipelines** using **Python**, **PySpark**, **Pandas**, and **T-SQL** to extract data from **Oracle** databases, transform it, and load it into semi-structured data in a **JSON-based data lake** layer and Microsoft SQL Server data warehouse, processing **5GB+** of data per pipeline run.
- Developed a reusable ETL validation framework using **Python** and **PySpark**, enabling automated validation of large datasets (**20M+ records**) across dev, uat, and prod environments, reducing data discrepancies by **60%**.
- Managed **MSSQL** database ETL releases and coordinated version-controlled deployment scripts, ensuring traceability, and rollback capability
- Assisted with **database monitoring and maintenance**, including performance diagnostics, log analysis, and troubleshooting database issues across enterprise systems.
- Executed enterprise database migration by upgrading **SQL Server 2019 to SQL Server 2022** across **DEVTEST, UAT, and PROD** environments, migrating 50M+ records, database schemas, stored procedures, triggers, and security configurations with minimal downtime.
- Built internal **data analytics dashboards** using **Python, T-SQL, and PowerBI** to surface database health, release versions, and database operational metrics for directorate and senior management.
- Worked in a cross-functional **Agile team** collaborating with Network engineers, Clients, and vendors to improve infrastructure readiness and support modernization initiatives.

Full Stack Developer Intern September 2023 - April 2024
University of Ottawa Ottawa, ON

- Supported development and maintenance of **Microsoft Power Platform applications**, including **Power Automate**, to enhance workflow automation, data processing, and operational efficiency.
- Developed **Power Platform solutions** including **Canvas Apps and Model-Driven Apps** integrated with **C# backend services** to support event management, user interactions, and data-driven workflows
- Implemented automated workflows using **Power Automate**, leveraging **HTTP triggers, scheduled flows, and API integrations** to enable real-time notifications, approvals, and data synchronization across systems
- Designed end-to-end integrations between **Power Apps and REST APIs**, enabling seamless communication between frontend applications and backend services
- Redesigned the University of Ottawa's emergency alert website with **C# (.Net Core)** and **Angular**, introducing a modern, interactive UI. Used by over **20,000** students daily. website: ue.uottawa.ca/
- Optimized existing **C#** REST APIs by reducing **SQL** query execution time for event data lookup by **30 %** and lines of code by **40 %**
- Participated in end-to-end **Software Development Lifecycle (SDLC)** processes, from requirements gathering and system design to implementation, testing, and production deployment

Cloud Developer Intern May 2023 - August 2023
Trend Micro Inc. Ottawa, ON

- Utilized **AWS Lambda** to monitor client files, images and code uploaded to the cloud and scan for vulnerabilities using Anchore APIs and Grype
- Programmed robust error handling for **AWS Lambda** functions in GoLang using goroutines to ensure proper timeout and recovery, reducing data corruption and improving service reliability

Cloud Developer Intern September 2022 - December 2022
Trend Micro Inc. Ottawa, ON

- Developed a serverless automation solution using **AWS Lambda, S3, and Athena crawlers** to trigger metadata updates on data changes, significantly reducing manual intervention and operational overhead
- Configured **Splunk** and **AWS CloudWatch** alarms for proactive monitoring of test logs and crashes, enhancing system reliability and reporting.
- Wrote **4 integration tests** using docker to test the AWS lambda and **3 unit tests** for the new features implemented